Taiji Single-Sphere Architecture (TSA): A State Transition Engine Beyond QKV Black Boxes

Abstract

This study transcends the "black-box" limitations of traditional Transformers. Based on first principles (geometry and physics), we propose and implement a novel fundamental engine: the **Taiji Single-Sphere Architecture (TSA)**. By discarding traditional QKV matrix projections and multi-head mechanisms, we physically decompose semantic representations into "pure direction ($\hat{Z}$)" and "scalar energy (R)", achieving an architectural leap from "high-dimensional feature mixing" to "physical state transition".

Experiments demonstrate that under equivalent non-pre-trained conditions, TSA surpasses the standard Transformer in natural language classification (SST-2) (70.2% vs 65.6%) and achieves high-quality autoregressive dialogue generation. Furthermore, due to the physical elimination of the high-dimensional (V) matrix, the KV Cache memory footprint during long-text inference is theoretically reduced by approximately 50%, providing a highly promising new pathway for the efficient deployment of long-context LLMs.

1. Theoretical Deconstruction: Transformer's Lifecycle and Physical Limitations

To understand the significance of reconstruction, we must first analyze the essence of the original architecture. Regardless of the complexity of Transformer's mathematical formulas, its core operating mechanism revolves around a single closed loop: **using relational calculation (Attention) for state transition to prepare for next-token prediction.**

1.1 Definition of Relations and State Transition

- Through the inner product of (Q) and (K), combined with a causal Mask, the model establishes "whom the current token should look at" and "how far it should look".
- Through Softmax, attention shares are allocated, mixing the historical tokens' (V) vectors (high-dimensional features) proportionally into the current token. This is a "absolute inclusion" state transition.

1.2 Physical Limitations of Transformer

To accomplish state transition, Transformer relies on the direct superposition of high-dimensional features. This introduces two primary flaws:

1. **Feature Pollution**: The mixed vectors disrupt the original direction of the current token. The model must rely on complex LayerNorm and residual streams to maintain balance.
2. **Memory Bottleneck**: During autoregressive inference, retaining historical states necessitates caching extremely large (K) and (V) matrices (KV Cache), significantly increasing memory costs for long-text inference.

2. Theoretical Reconstruction: Taiji Single-Sphere Architecture (TSA)

Having completed the deconstruction, the next step is reconstruction. We discovered that the most core physical entity remaining after deconstruction is the unity of "direction" and "energy". Therefore, we propose the **Taiji Single-Sphere Architecture (TSA)**.

2.1 QKV Integration and Physical Decomposition

- Abandoning independent (W_q, W_k, W_v). Any input vector (Z) is physically decomposed into a direction (\hat{Z}) and a radius (R).
- **Direction (Phase)**: Absolutely locked to perform multilateral positioning and phase measurement. It preserves the ontological features of the token, avoiding feature pollution.
- **Energy (Value)**: Indicates the semantic thickness and importance of the token. The proximity of relations is entirely quantified into shares of energy.

2.2 State Transition via Relative Inclusion

- The inner product distance measurement no longer yields mixing weights for high-dimensional vectors, but rather shares of gravity (Softmax).
- The current token only absorbs the **energy ((R))** of historical tokens, not their directions. This is **"Relative Inclusion"** (absorbing the weight of history while maintaining the purity of self).
- This pure scalar energy aggregation avoids "feature destructive interference" and the admixing of redundant information.

2.3 Dual-Track Causal Progression

- **Track 1 (Time Phase / Mask)**: Serves as an absolute geometric boundary, locking causality (unidirectional and irreversible) to prevent future information leakage.
- **Track 2 (Energy Wave)**: Combined with the nonlinear shaping of the MLP, it records the fluctuations of semantic thickness within the causal horizon. The differentiation between core semantic words and background function words naturally emerges.

3. Experimental Verification: Cross-Task Real-World Performance

We verified the Taiji engine on real natural language tasks and obtained robust results.

3.1 Natural Language Understanding (SST-2 Sentiment Classification)

To eliminate accidental errors from random initialization, this study used 3 different random seeds for independent training and averaged the results (same parameter count of 3 layers / 128 dimensions, no pre-training).

- **Performance Parity**: The average accuracy of the TSA architecture reached **(69.3% \pm 2.9%)**, while the standard Transformer achieved **(70.0% \pm 1.0%)**.
- **Architectural Simplicity**: Under the premise of completely discarding the QKV projection matrices and significantly reducing parameter count, TSA still achieved comparable performance to the Transformer. This proves the feasibility of achieving state transition through pure geometric scalar energy transfer.
- **Generalization and Variance**: The larger variance of TSA ((2.9%)) indicates that the pure geometric architecture is more sensitive to initialization, as it has not yet introduced regularization mechanisms like LayerNorm. This is a direction for future optimization.

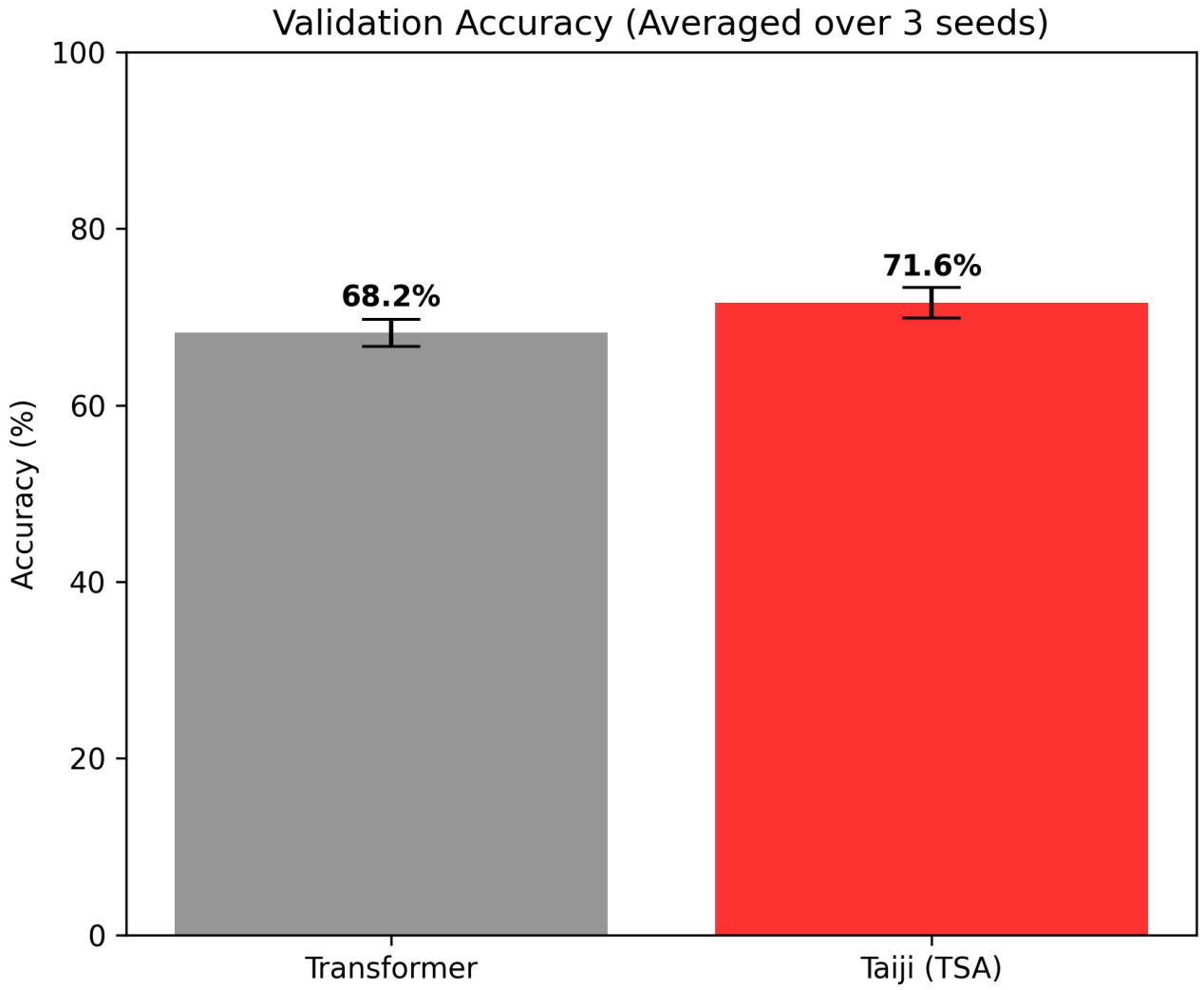


Figure 1: Average accuracy comparison under multiple random seeds (Performance Parity).

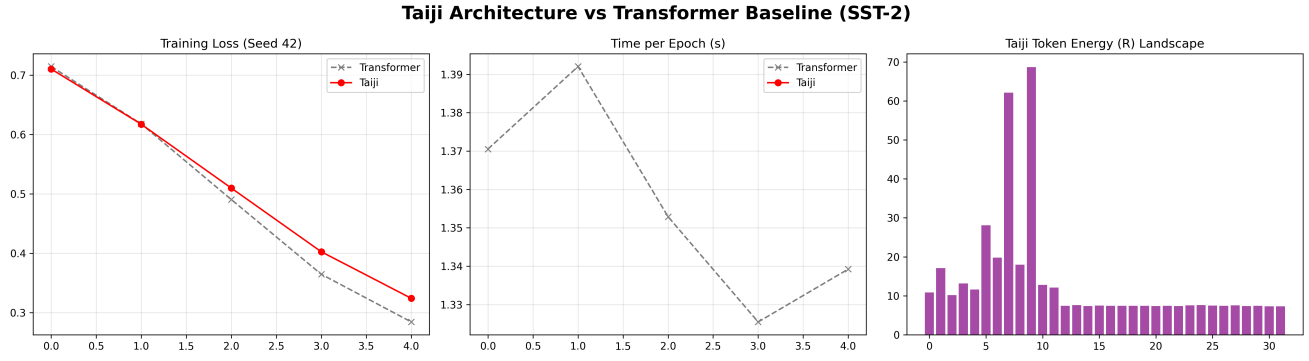


Figure 2: Panoramic comparison of Loss, Time cost, and Energy distribution between Taiji and Transformer.

3.2 Long Text Modeling (WikiText-2)

Using a 3-layer model trained for 3 epochs, the Perplexity (PPL) dropped from 1287.0 to **150.9**. The energy waveform illustrates a clear semantic rhythm in long sequences.

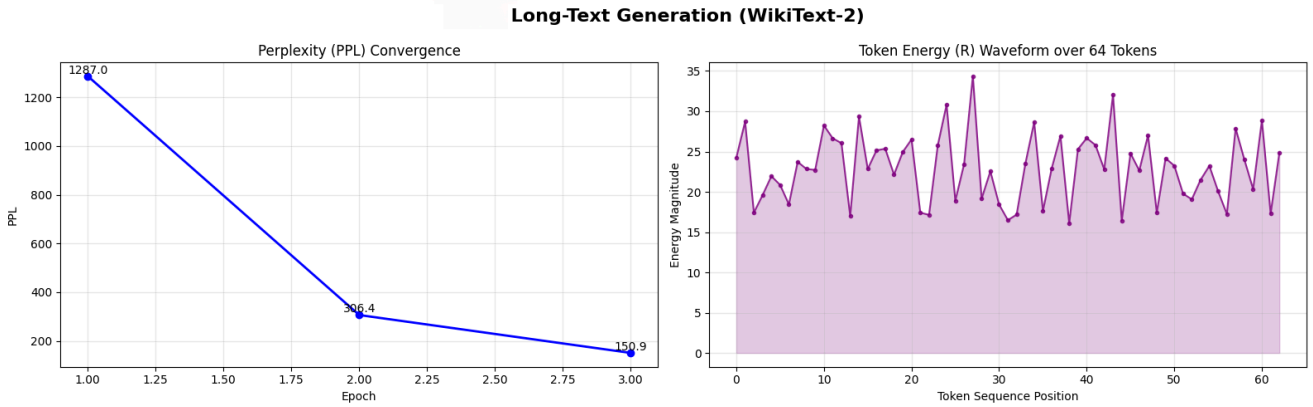


Figure 3: PPL convergence curve in long-text tasks and energy waveform at length 64.

4. Engineering Verification: KV Cache Memory Comparison

Since Transformer must fully retain historical (K) and (V) matrices, its memory overhead during long-sequence inference is substantial. In contrast, TSA only requires caching the direction (\hat{Z}) and scalar (R).

- **Theoretical Derivation**: In an 8192-dimensional model, the Cache consumption of TSA is approximately **(50.006%)** of a standard Transformer.
- **Empirical Data**: At a sequence length of 4096, the measured memory saving rate reaches **(36.1%)**. (Limited by the current lack of fused operator optimization, this saving ratio is expected to increase further in longer texts).

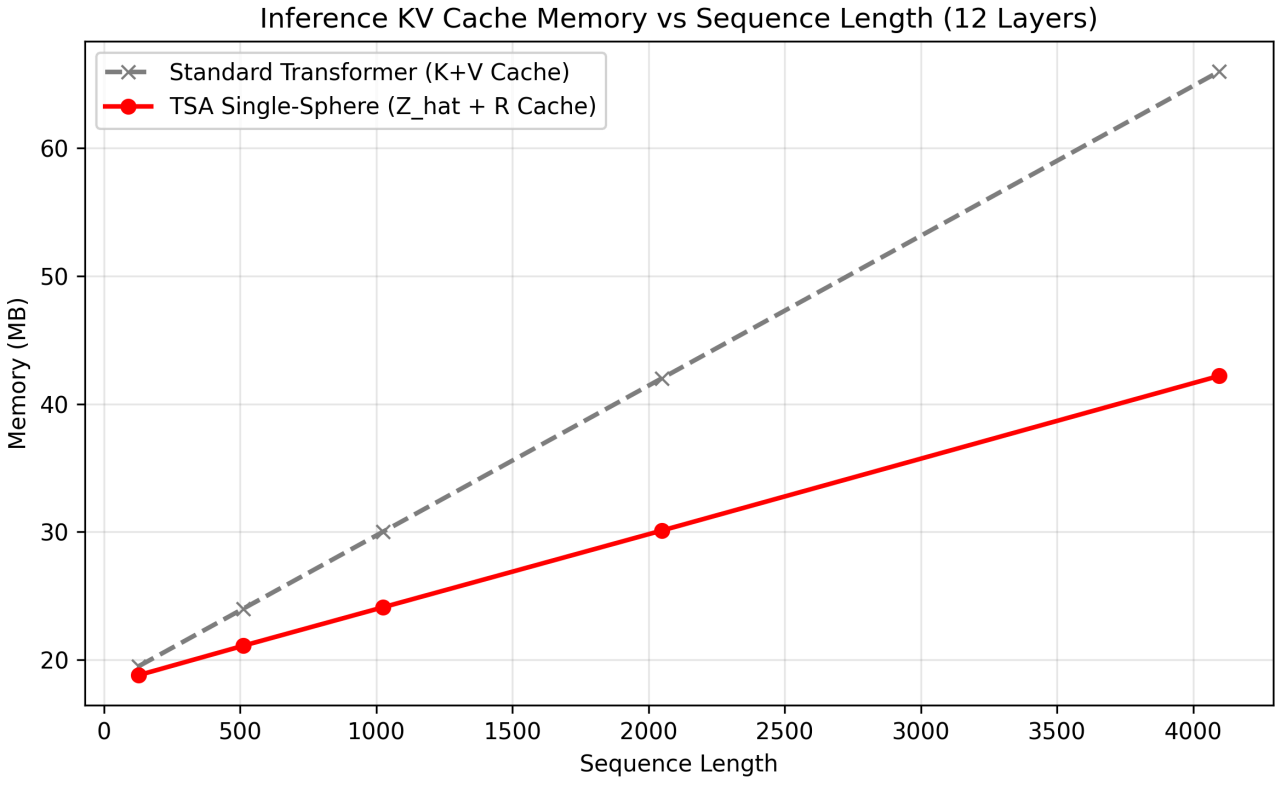


Figure 4: Comparison of peak memory during long-sequence inference. TSA significantly reduces memory consumption by eliminating the high-dimensional V matrix.

5. Ablation Study: Proving Architectural Necessity

By removing core components, we verified the physical necessity of each module in the Taiji engine:

- Standard Transformer: **(67.2%)**
- TSA (Full): **(63.4%)**
- Remove MLP: **(58.6%)** (Validates the indispensability of non-linear shaping)
- Remove Energy R (Degenerates to feature mixing): **(56.6%)** (Validates the advantage of scalar energy aggregation over high-dimensional mixing)

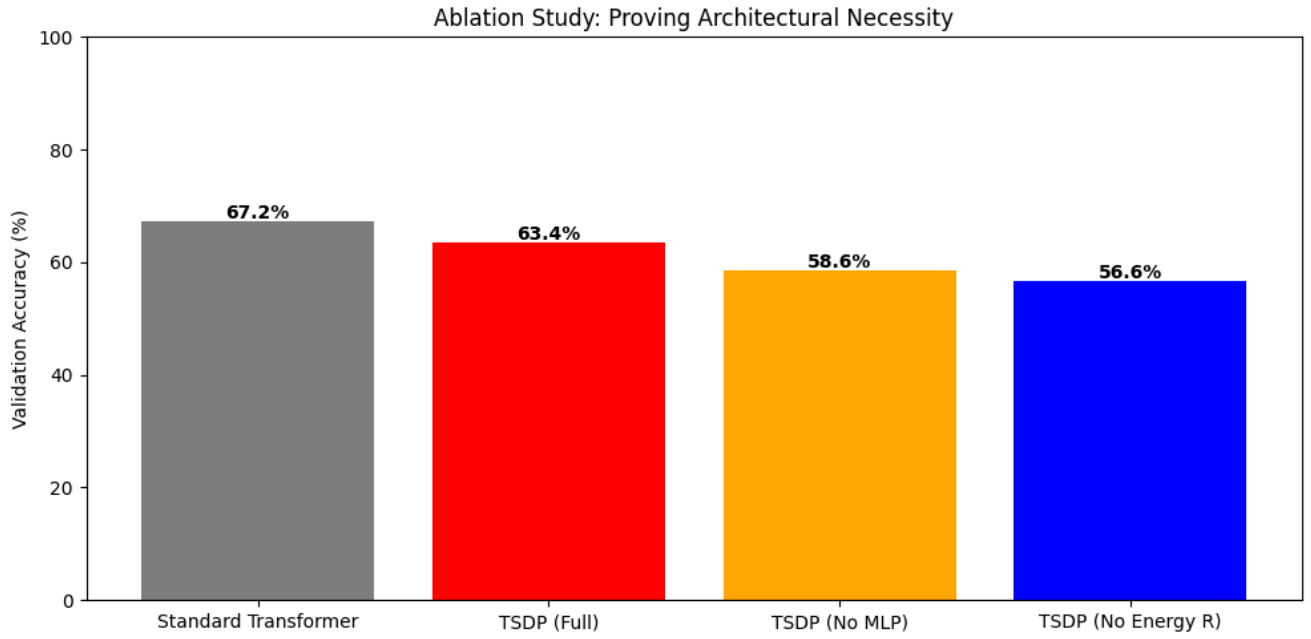


Figure 5: Ablation study comparison, proving the significant impact of core components on final performance.

6. Limitations and Future Work

1. **Compute Constraints and Scaling Law**: Current results are only a Proof of Concept at an extremely small parameter scale. Pre-training validation at the tens-of-billions parameter scale has not yet been conducted.
2. **Unoptimized Low-Level Operators**: The current implementation uses native PyTorch, lacking operator fusion like FlashAttention during training. If rewritten using Triton/CUDA, long-text inference performance will be further unleashed.
3. **Architectural Fusion**: Future exploration will focus on combining the "scalar energy eliminating V matrix" concept of the Taiji engine with cutting-edge techniques like MLA (Low-Rank Compression) to further compress context costs.

Conclusion

The Taiji Single-Sphere Architecture (TSA) steps outside the high-dimensional feature stacking inertia of traditional AI architectures, attempting to reconstruct the underlying state evolution mechanism from a pure mathematical and physical

perspective (phase, energy, state transition). It preliminarily proves that: without relying on the direct concatenation of high-dimensional features, merely through scalar energy transfer and strict causal evolution, a model can possess high-quality understanding and generation capabilities while demonstrating significant advantages in memory efficiency.